



Artificial Intelligence (CSE3013)

AI: Applications and Evolution

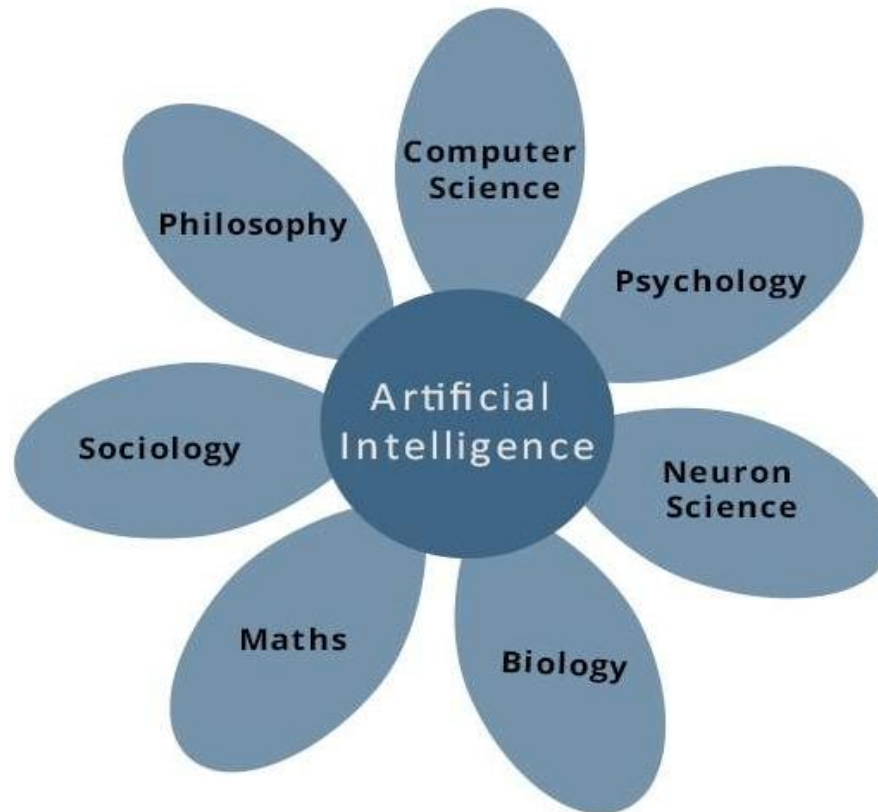
-Nitin Singh Rajput

Outline

- ▶ Applications and Domains of AI
- ▶ Evolution of AI

Components of AI

- ▶ Artificial Intelligence is not an easy science to describe
- ▶ It is a fuzzy borders with the following disciplines:





The Foundation of AI

► ***Philosophy***

- The study of human intelligence began with no formal expression
- Initiate the idea of mind as a machine and its internal operations

The Foundation of AI

- ▶ **Mathematics** formalizes the three main area of AI:
 1. Computation
 2. Logic or Decision Theory
 3. Probability

The Foundation of AI

- ▶ *Computation* leads to analysis of the problems that can be computed
 - ▶ complexity theory
- ▶ *Probability* contributes the “degree of belief” to handle uncertainty in AI
- ▶ *Decision theory* combines probability theory and utility theory

The Foundation of AI

► **Psychology**

- How do humans think and act?
- The study of human reasoning.
- Provides reasoning models for AI
- Strengthen the ideas
 - humans and other animals can be considered as information processing machines



The Foundation of AI

▶ **Computer Engineering**

- ▶ How to build an efficient computer?
- ▶ Provides the artifact that makes AI application possible
- ▶ The power of computer makes computation of large and difficult problems more easily
- ▶ AI has also contributed its own work to computer science



The Foundation of AI

▶ **Control theory and Cybernetics**

- ▶ How can artifacts operate under their own control?
- ▶ The artifacts adjust their actions
 - ▶ To do better for the environment over time
 - ▶ Based on an objective function and feedback from the environment
- ▶ Not limited only to linear systems but also other problems
 - ▶ language, vision, and planning, etc.

The Foundation of AI

▶ **Linguistics**

- ▶ For understanding natural languages
 - ▶ different approaches has been adopted from the linguistic work
- ▶ Formal languages
- ▶ Syntactic and semantic analysis
- ▶ Knowledge representation

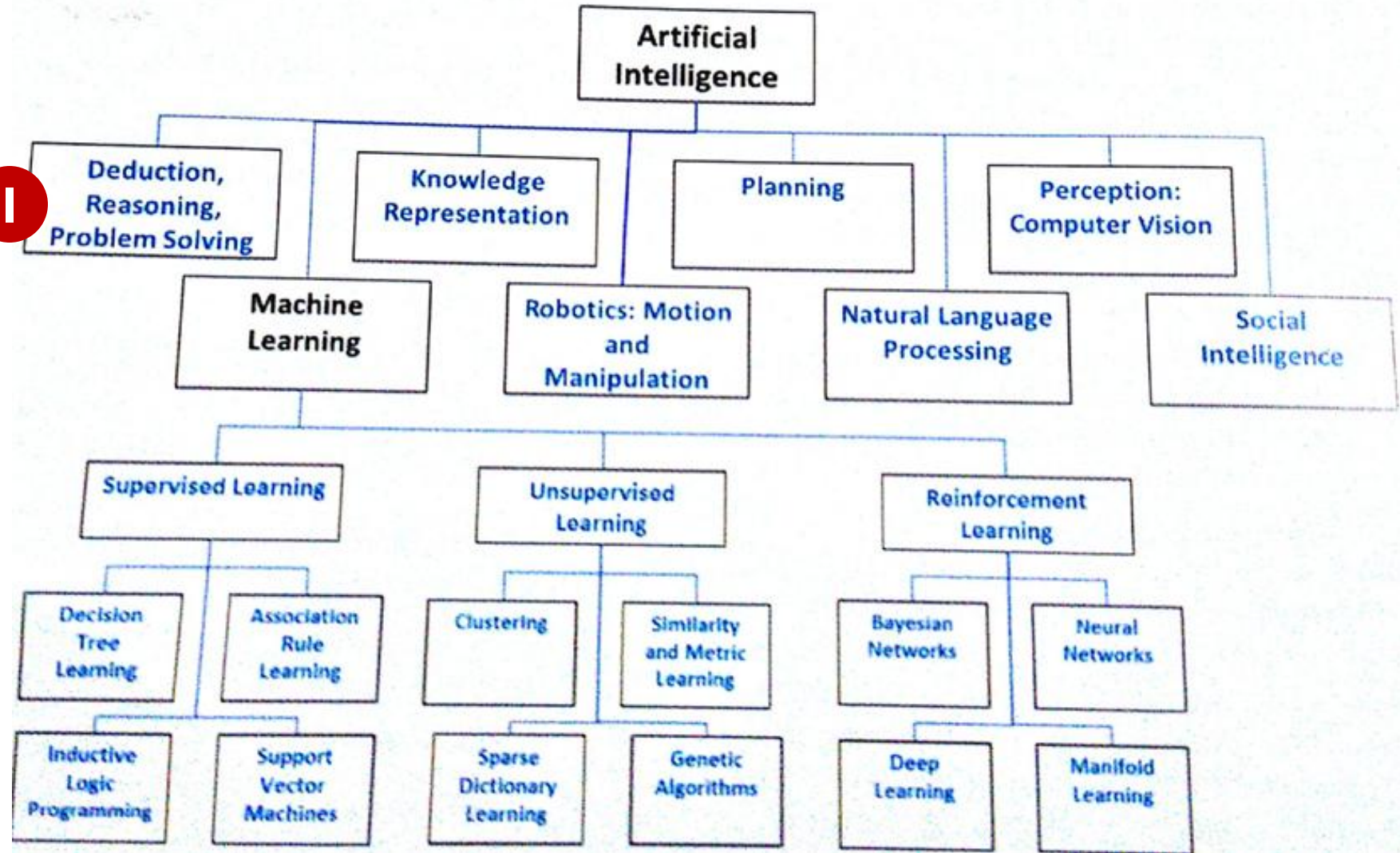
▶ **Biology and Neuron-Science**



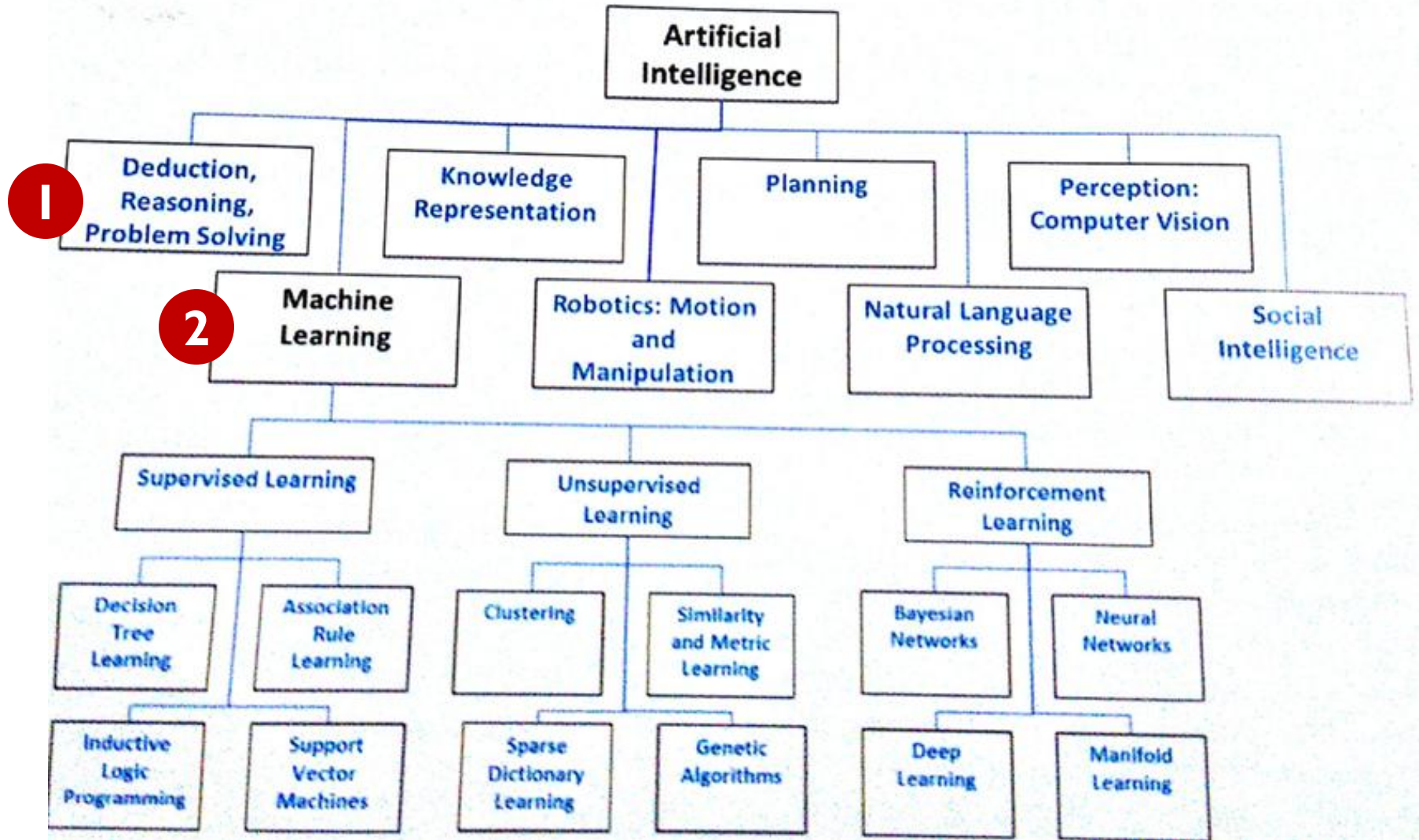
Some Advantages of Artificial Intelligence

- ▶ More powerful and more useful computers
- ▶ **New and improved interfaces**
- ▶ Solving new problems
- ▶ **Better handling of information**
- ▶ Relieves information overload
- ▶ **Conversion of information into knowledge**

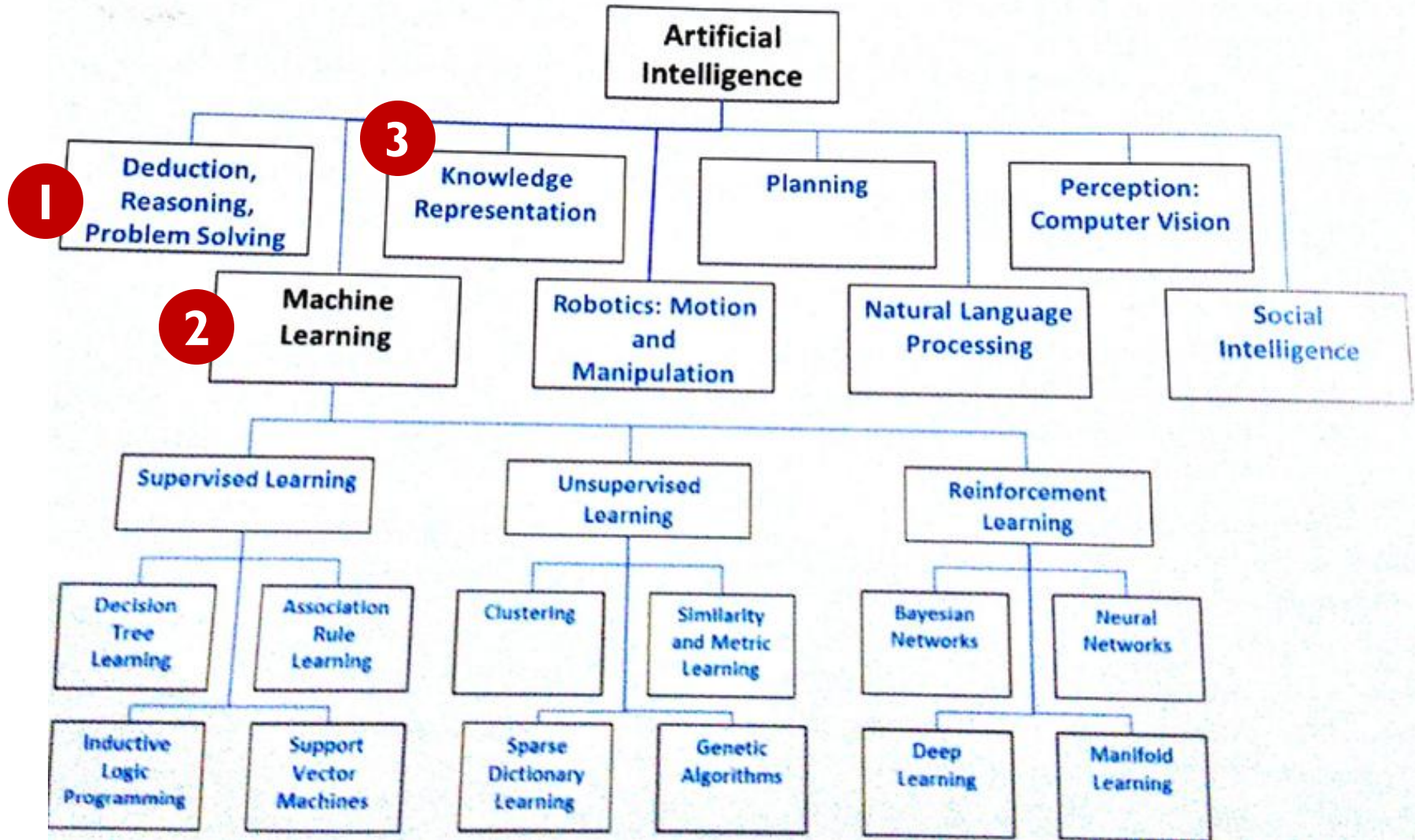
Fields of AI



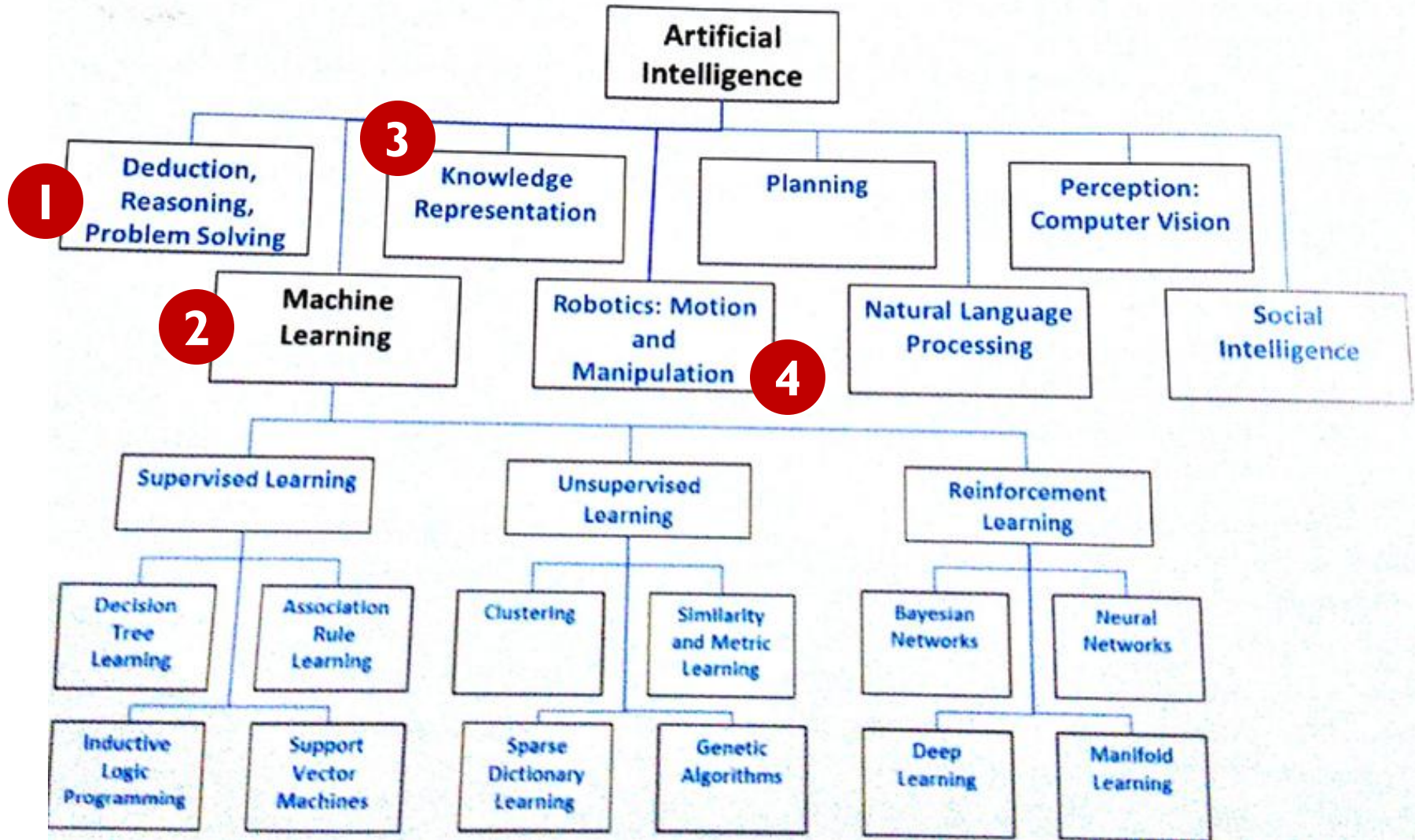
Fields of AI



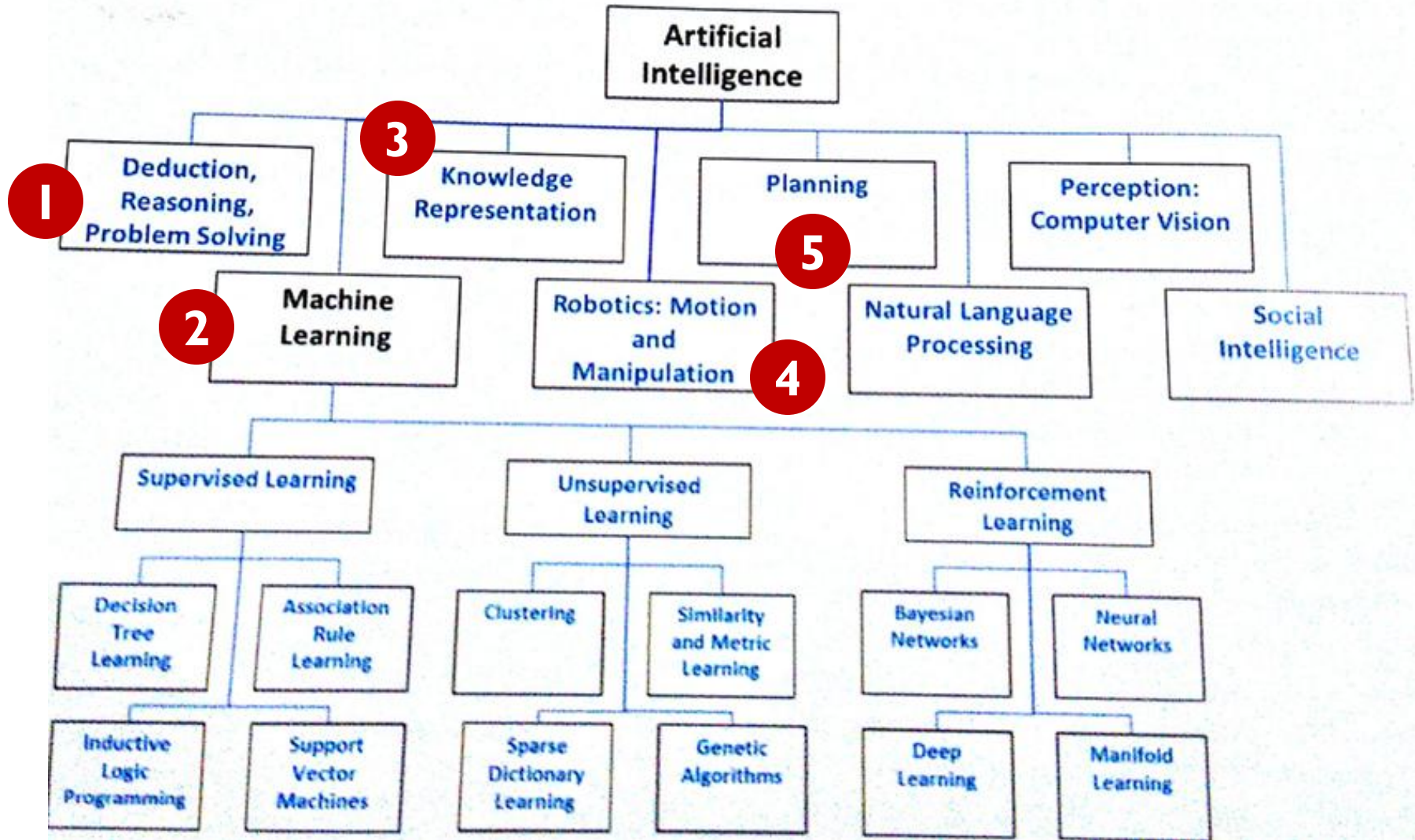
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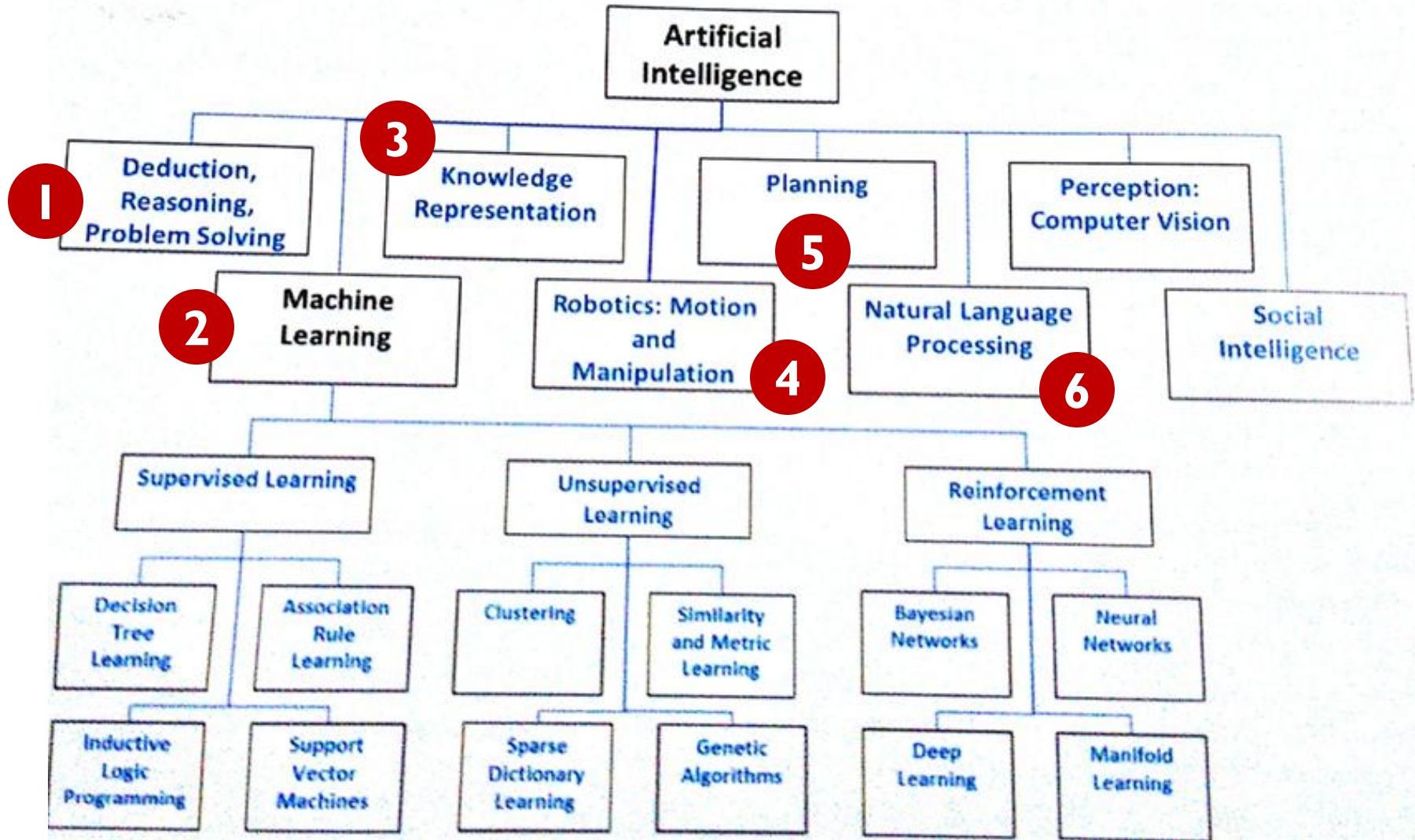
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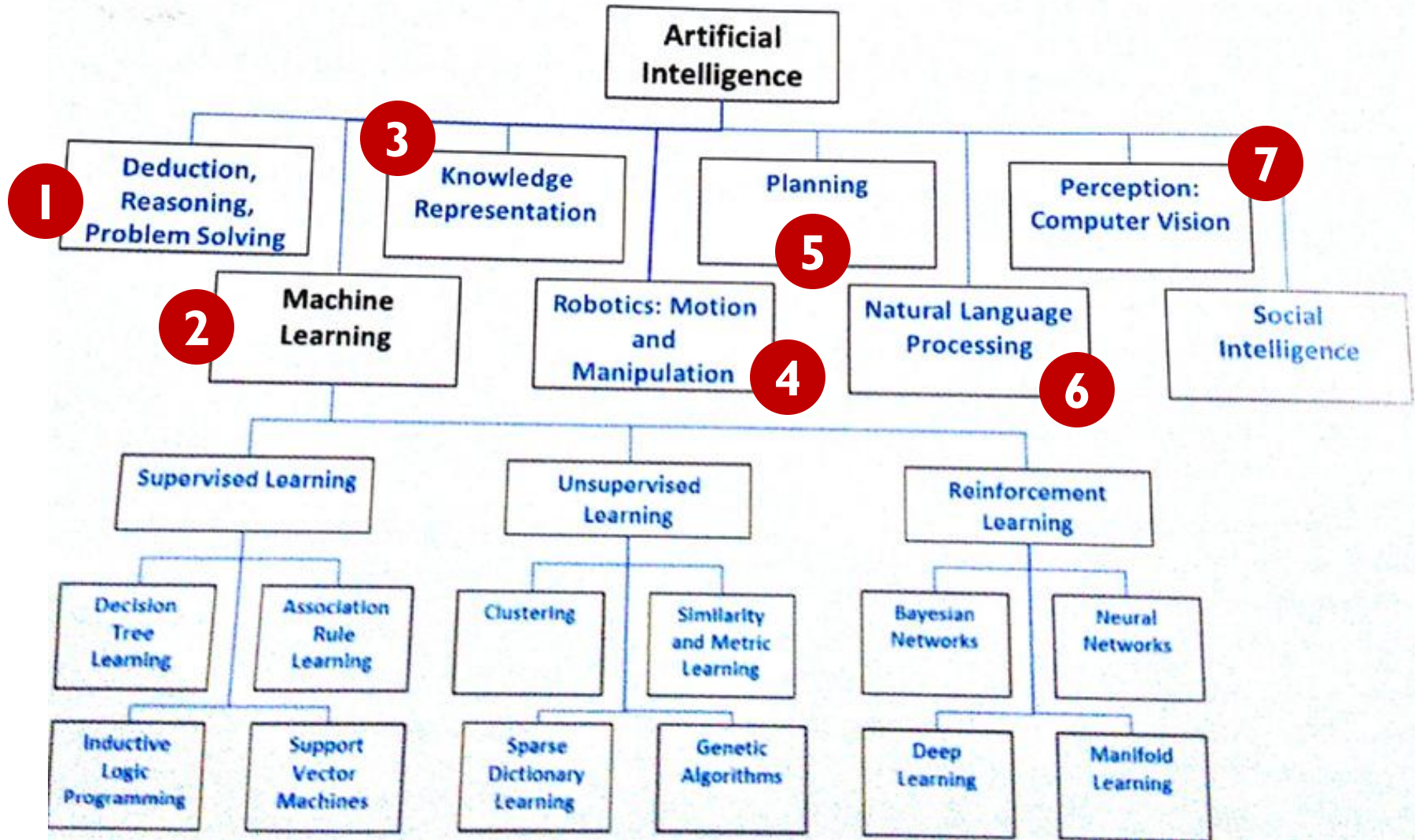
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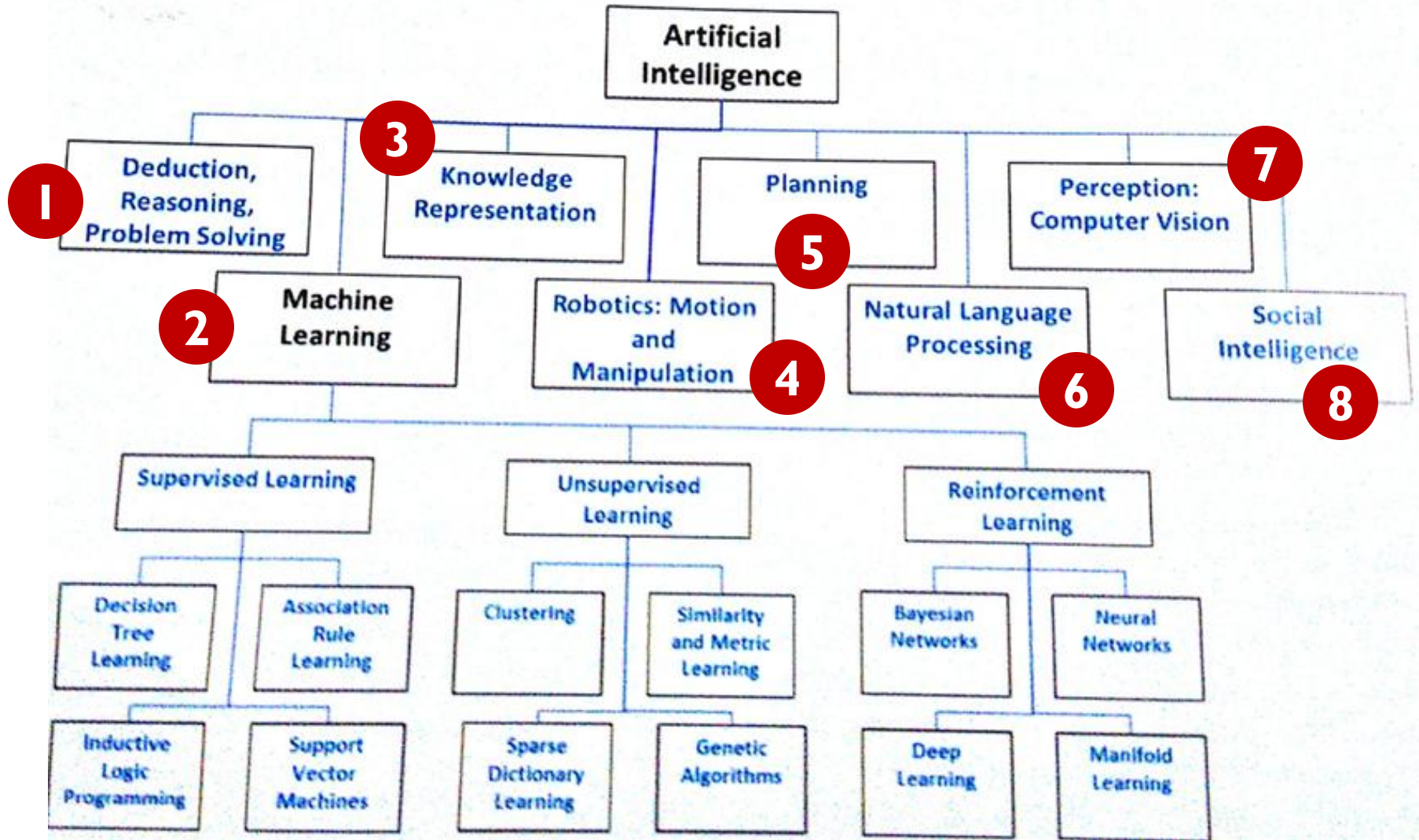
Fields of AI



Fields of AI



Fields of AI



Goals of AI

- ▶ To make computers more useful by letting them take over dangerous, complex or tedious tasks from human
- ▶ Understand principles of human intelligence

Backlogs in the Development of Artificial Intelligence

- ▶ Increased costs
- ▶ Difficulty with software development - *slow & expensive*
- ▶ Few experienced programmers
- ▶ Few practical products have reached the market as yet.

Generic Techniques Developed for AI

- ▶ Forward/backward chaining (reasoning)
- ▶ Resolution theorem proving (reasoning)
- ▶ Proof planning (reasoning)
- ▶ Constraint satisfaction (reasoning)
- ▶ Davis-Putnam method (reasoning)
- ▶ Minimax search (games)
- ▶ Alpha-Beta pruning (games)
- ▶ Case-based reasoning (expert systems)
- ▶ Knowledge elicitation (expert systems)

Generic Techniques Developed for AI

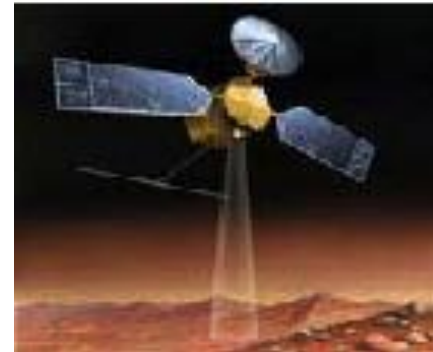
- ▶ Neural networks (learning)
- ▶ Bayesian methods (learning)
- Explanation based (learning)
- Inductive logic programming (learning)
- Reinforcement (learning)
- Genetic algorithms (learning)
- Genetic programming (learning)
- Strips (planning)
- N-grams (NLP)
- Parsing (NLP)
- Behaviour based (robotics)
- Cell decomposition (robotics)

Application Domains

- ▶ Agriculture
- ▶ **Architecture**
- ▶ Art
- ▶ **Astronomy**
- ▶ Bioinformatics
- ▶ **Email classification**
- ▶ Engineering
- ▶ **Finance**
- ▶ Fraud detection
- ▶ **Information retrieval**
- ▶ Law
- ▶ Mathematics
- ▶ **Military**
- ▶ Music
- ▶ **Scientific discovery**
- ▶ Story writing
- ▶ **Telecommunications**
- ▶ Telephone services
- ▶ **Transportation**
- ▶ Tutoring systems
- ▶ **Video games**
- ▶ Web search engines

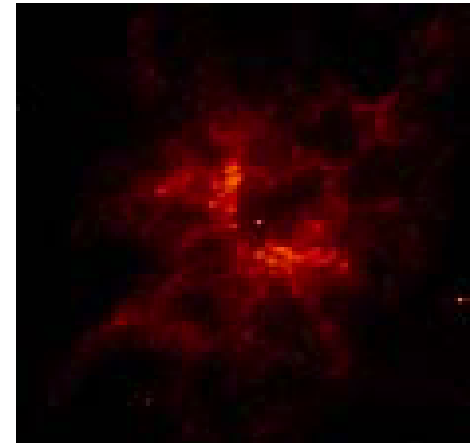
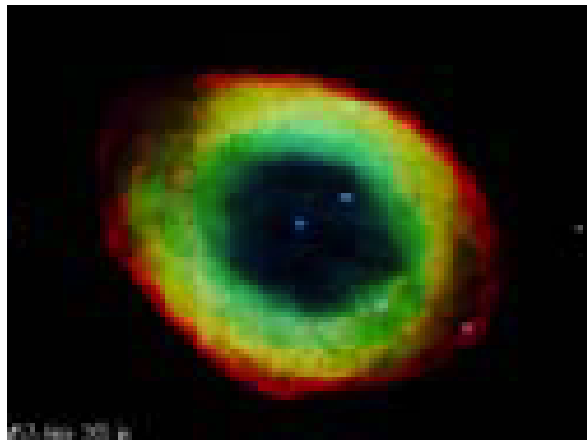
Application of AI

- ▶ Autonomous Planning & Scheduling:
 - ▶ Autonomous rovers



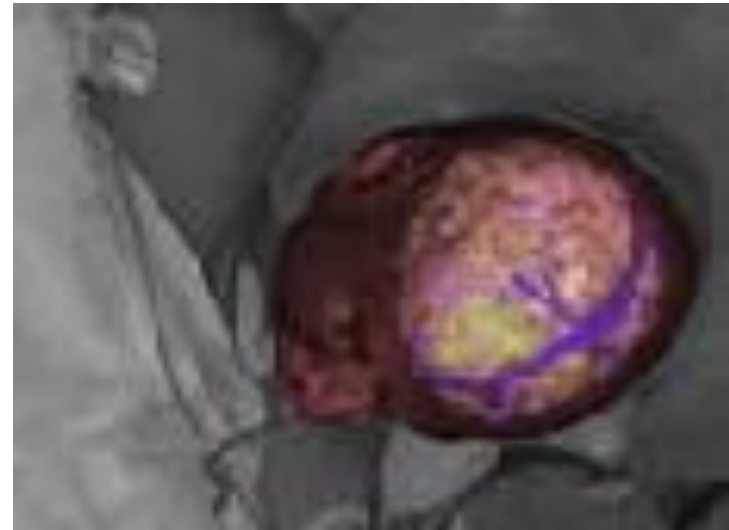
Application of AI

- ▶ Autonomous Planning & Scheduling:
 - ▶ Analysis of data



Application of AI

- ▶ Medicine:
 - ▶ Image guided surgery



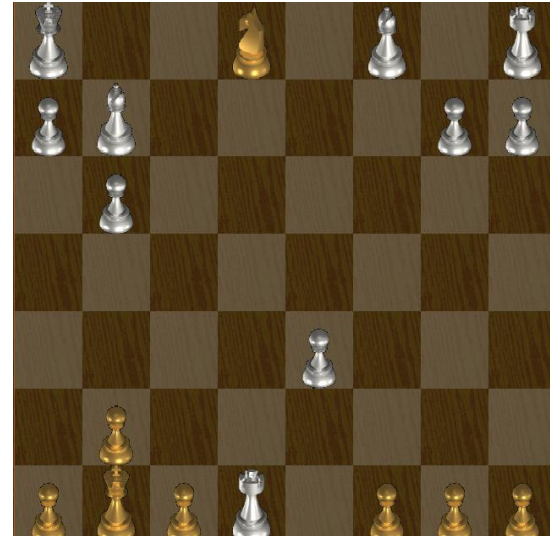
Application of AI

- ▶ **Transportation:**
 - ▶ Autonomous vehicle control
 - ▶ **Pedestrian detection:**



Application of AI

Games:



Application of AI

► Robotic toys:

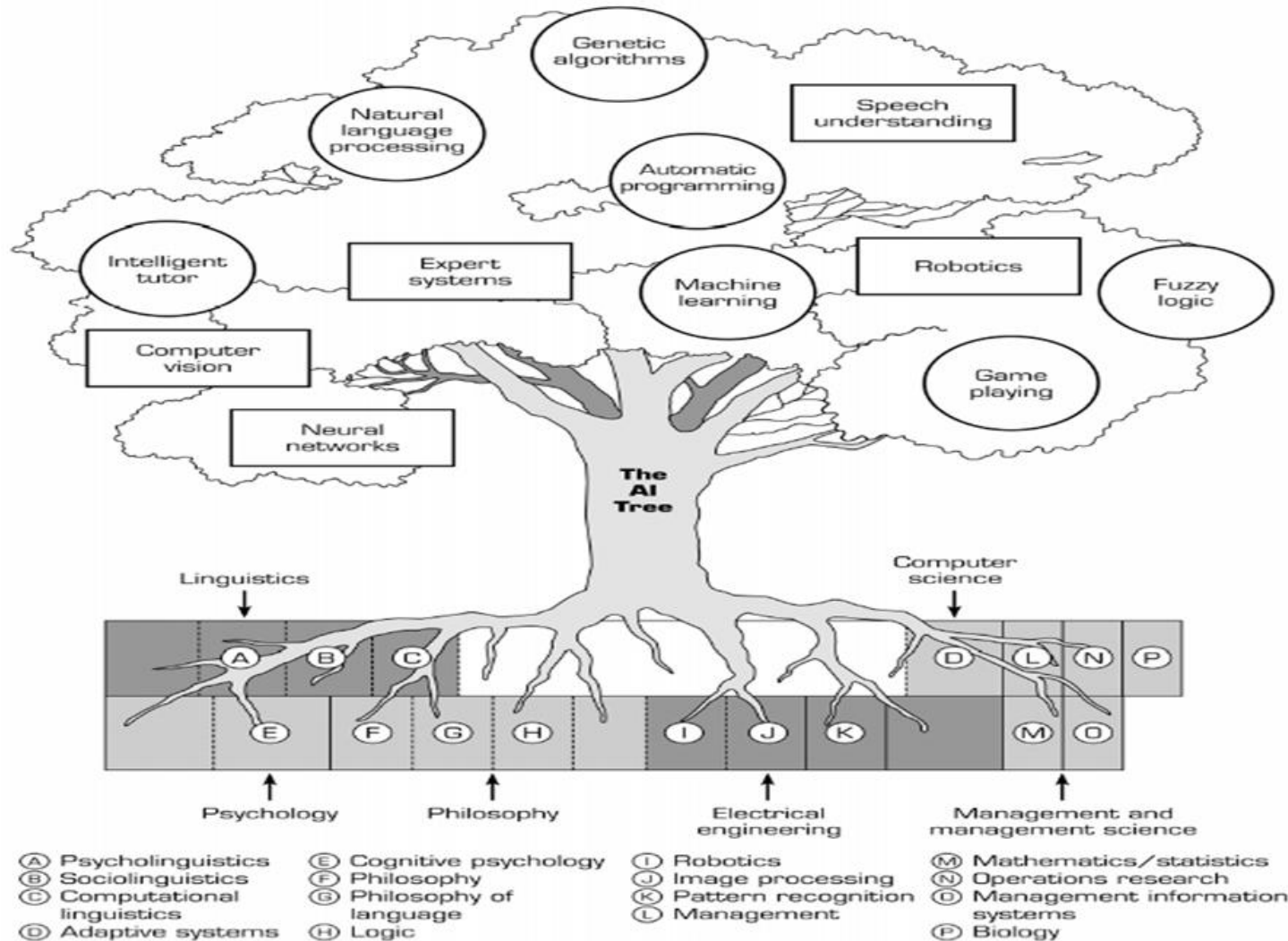




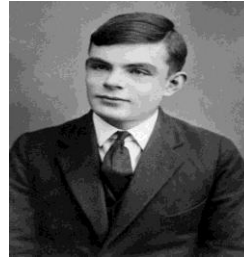
Application of AI

- ▶ Other application areas:
 - ▶ **Bioinformatics:**
 - ▶ Gene expression data analysis
 - ▶ Prediction of protein structure
 - ▶ **Text classification, document sorting:**
 - ▶ Web pages, e-mails
 - ▶ Articles in the news
 - ▶ **Video, image classification**
 - ▶ **Music composition, picture drawing**

Disciplines & Applications of Artificial Intelligence



Evolution of AI

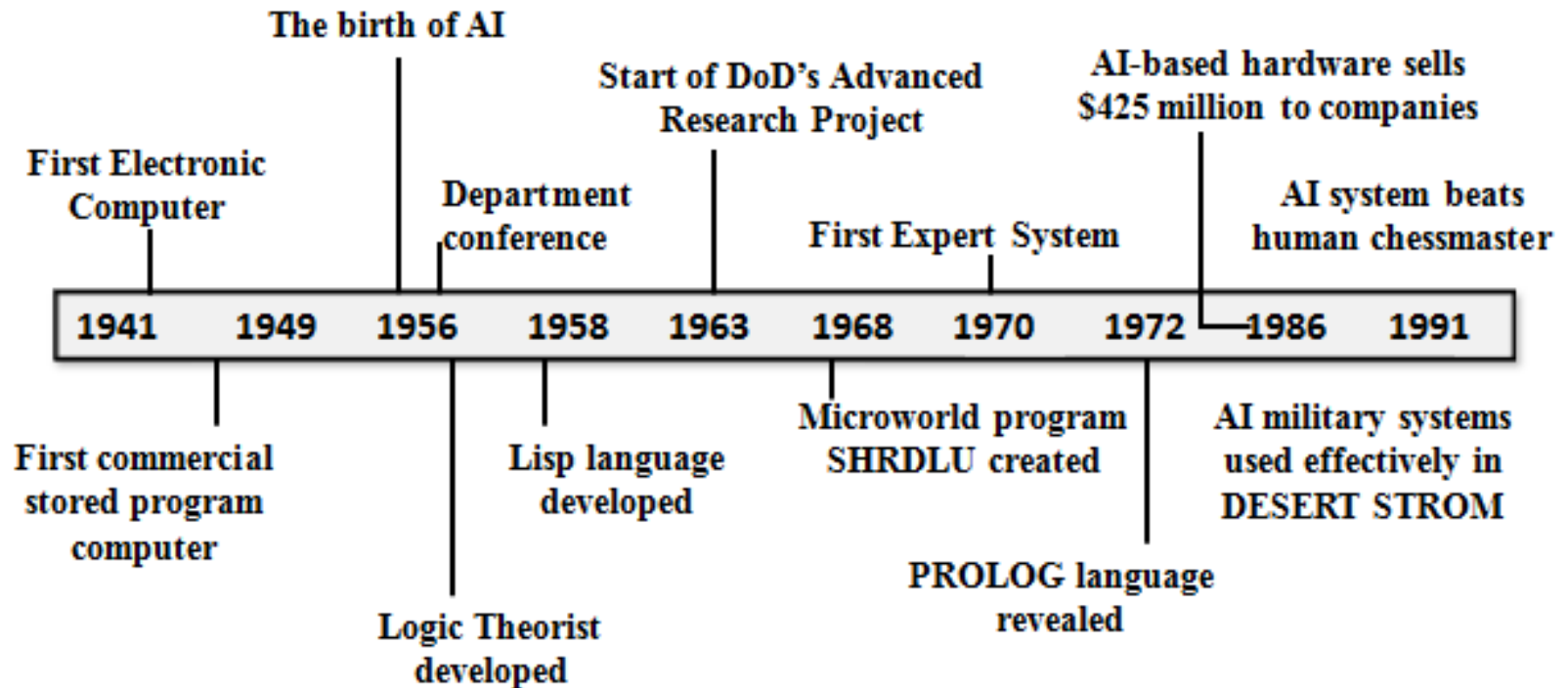


- ▶ The origin of AI can be seen in Turing's work in his paper on **intelligent machines** published in 1950.



- ▶ John McCarthy introduce the word **AI** in 1956

Timeline for Major AI Events

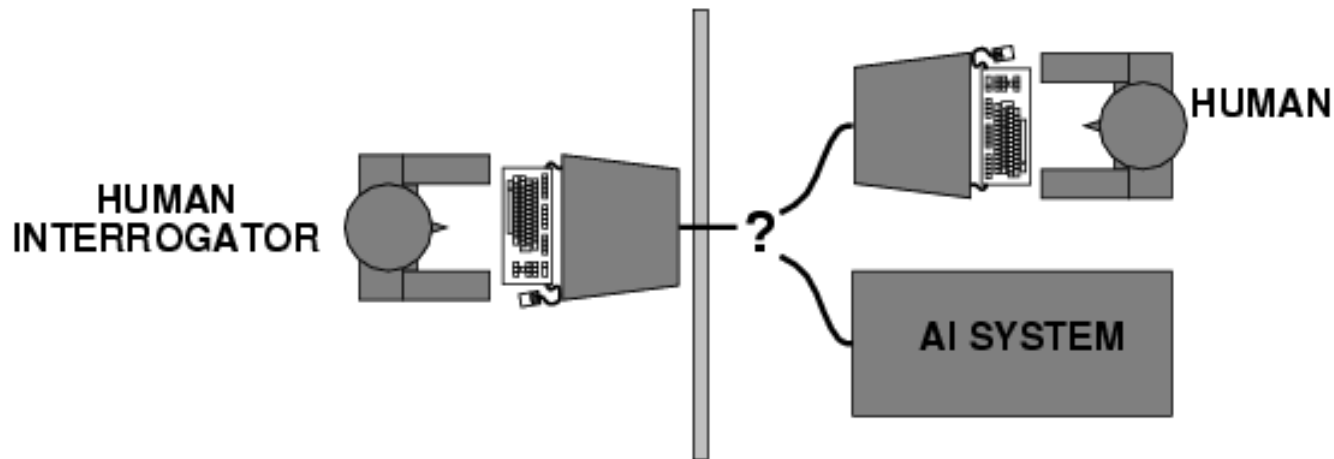


The Turing Test

- ▶ Turing (1950):
- ▶ "Can machines think?" → "Can machines behave intelligently?"
- ▶ Designed Operational test for intelligent behavior:
 - ***Imitation Game***

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AI based Systems

***Expert Systems –
Apply expert
knowledge to
difficult,
real world problems.***

***Knowledge Based Systems –
Make domain knowledge explicit***

***AI Programs-
Exhibit intelligent behavior by skillful
application of heuristics***



About the Course

Any Query ???



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